

Name _____ Baggett/Fall2026/207/Quiz1
first name last name

No Calculators.

Questions 1 and 2. Circle **True** or **False**. Then, in **one sentence**, say why — if the statement is false, the fastest way is usually to give an example that breaks it. **1 point for the circle, 2 for the reason.**

1. (3 pts) The average rate of change of f on $[a, b]$ is the slope of the secant line through $(a, f(a))$ and $(b, f(b))$.

☒ **True** ☐ **False**

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Why? **That is the definition:** $\frac{f(b)-f(a)}{b-a}$ **is exactly the secant slope.**

2. (3 pts) If $\lim_{x \rightarrow 2} f(x) = 5$, then $f(2) = 5$.

☐ **True** ☒ **False**

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Why? **The limit ignores the value at the point. f need not even be defined at 2.**

3. (4 pts) Let $f(x) = 3 - x^2$. Form the difference quotient at $a = 2$ and simplify it as far as it will go:

$$\frac{f(2+h) - f(2)}{h}$$

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$$f(2+h) = 3 - (2+h)^2 = 3 - 4 - 4h - h^2 = -1 - 4h - h^2, \quad f(2) = -1.$$

$$\frac{(-1 - 4h - h^2) - (-1)}{h} = \frac{-4h - h^2}{h} = -4 - h$$

4. (4 pts) A drone rises straight up. Its height above the ground after t seconds is

$$s(t) = 40t - 5t^2 \text{ metres}, \quad 0 \leq t \leq 8.$$

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- (a) Find the drone's **average velocity** over the interval $[1, 3]$. Include units.

$$s(3) = 75, s(1) = 35, \text{ so } \frac{75-35}{3-1} = 20 \text{ m/sec.}$$

- (b) In **one sentence**, say what that number tells you about the drone.

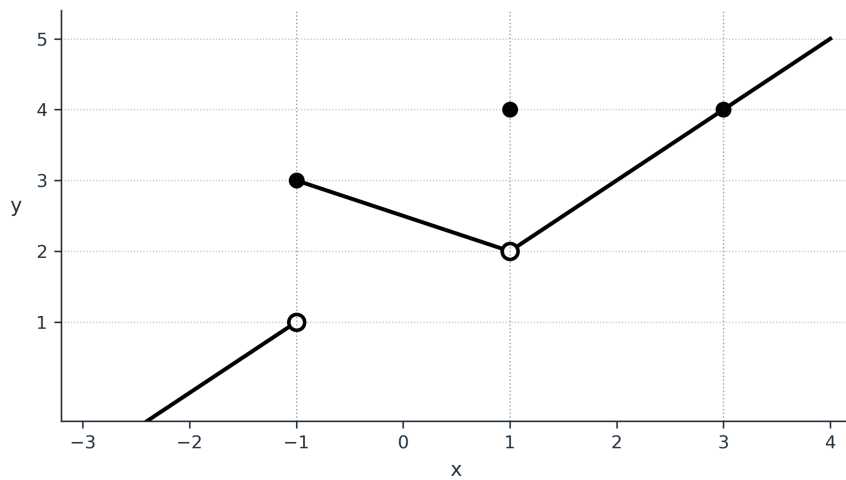
Over those two seconds the drone gained height at an average rate of 20 metres per second.

- (c) Over $[1, 1.5]$, $[1, 1.1]$ and $[1, 1.01]$ the average velocities are 27.5, 29.5 and 29.95 m/sec. What number are they heading for, and what does it say about the drone at the **instant** $t = 1$?

30 m/sec — the instantaneous velocity at $t = 1$.

5. (6 pts) Use the graph of $y = f(x)$ below. If a limit does not exist, say so.

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(a) $\lim_{x \rightarrow -1^-} f(x) = \underline{\quad 1 \quad}$ (b) $\lim_{x \rightarrow -1} f(x) = \underline{\quad \mathbf{DNE} \quad}$

(c) $\lim_{x \rightarrow 1} f(x) = \underline{\quad 2 \quad}$ (d) $f(1) = \underline{\quad 4 \quad}$

(e) $\lim_{x \rightarrow 3} f(x) = \underline{\quad 4 \quad}$ (f) $f(3) = \underline{\quad 4 \quad}$